Lab 5 report

R_D:

$$A\,CS\ = (-g_m{*}r\ o\ ){*}\frac{R\ D\ ||R\ L}{R\ out + R\ D\ ||R\ L} where R\,out = R\,S + r\ o + g\ m\ {*} r\ o\ {*} R\,S$$

since r_o goes to ∞, the R_{out} term grows higher than the parallel term.

$$A\ CS\ = (-g\ m\ {*} r\ o\ ) * \frac{R\ D\ ||R\ L}{g\ m\ {*} r\ o\ {*} R\ S}$$

Then the terms will cancel out from the denominator.

$$A\ CS\ = -\frac{R\ D\ ||R\ L}{R\ S}$$

solving: $R_D||R_L = 1k\ \Omega$ therefore $R_D = 2k\Omega$

R_S:

$$A_SF = \frac{R\ D\ ||R\ L}{R\ out + R\ D\ ||R\ L} = \frac{R\ D\ ||R\ L}{\frac{1}{g_m} + R\ D\ ||R\ L}$$

Solving: R_S || R_L = $\frac{1}{g_m}$ = 1kΩ therfore R_S = 2kΩ

I_D: Next, I_D can be solved using the equation

$$g_m = \sqrt{\mu C_o x (W/L) I_D} \rightarrow I_D = 1.502mA$$

V_{S1}: Given: $g_m = K_n * V_{ov} \rightarrow V_{ov} = \frac{g_m}{K_n} = 3.003V$

V_{GS}: $V_{GS} = V_{ov} + V_{th} = 5.003V$

V_G: $V_G = V_S + V_{GS} = 8.006V$

V_{D1}: $V_{S1} = 3.003V \rightarrow V_{D1} = V_{DD} - V_{S1} = 6.997V$

$V_{DS1} : V_{DS1} = V_{D1} - V_{S1} = 3.994V$

M1 Operating Region: The operating region is saturation with $V_{DS} > V_{ov} \rightarrow 3.994V > 3.003V$

Exercise 14

$$g_{m2} = 1.5mA/V$$

$$A_{CG} = 1V/V, where A_{CG} = \frac{v_{g1}}{v_{s2}}$$

$$R_{in} = 300\Omega$$

Derive an expresion for R_{in} and solve for R_2

$$R_{in} = R_2 || \frac{1}{g_{m2}} \rightarrow \frac{\frac{R_2}{g_{m2}}}{R_2 + \frac{1}{g_{m2}}} = 300\Omega \rightarrow R_2 = \frac{\frac{300}{g_{m2}}}{-300 + \frac{1}{g_{m2}}}$$

$$A_{CG} = g_{m2} R_3 \rightarrow R_3 = 667\Omega$$

$$I_D = \frac{g_{m2}^2}{2K_n} = 4.505V$$

$$V_{ov2} = \frac{g_{m2}}{K_n} = 4.505V$$

$$V_{GS2} = V_{ov2} + V_{th} = 6.505V$$

$$V_{S2} = I_{D2} R_2 = 1.84V$$

$$V_{G2} = V_{S2} + V_{GS2} = 8.34V$$

$$V_{D2} = V_{DD} - I_{D2} R_3 = 7.75V$$

$$V_{DS2} = V_{D2} - V_{S2} = 5.9V$$

The M2 operating region is saturation. $V_{DS2} > V_{ov2} \rightarrow 5.9V > 4.5V$

Simulation

Exercise 15

$$\Delta A = 20log10(\frac{A_{CS+A_{SF}}}{0.001})$$

$$\Delta\phi = 2\pi f \Delta t$$

Using the formulas above and the simulation in the appendix these values were calculated.

Parameter	Value
v_{o1}	255.93mV
v_{o2}	255.93mV
v_{in}	999.99mV
$t_o(v_{o1})$	9.9595μs
$t_o(v_{o1})$	9.9598μs
ΔA	54.18dBm
$\Delta\phi$	-83.252μrad

Viewing both the phase imbalance and the amplitude imbalance we can see that the circuit is indeed matched and outputs a differential output.

INSERT PLOTS AND TALK ABOUT IT

Physical Experiments

Procedure 1 and 2

Parameter	Design Prediction	Adjusted Predictions	Actual Measurements
R_1	100kΩ	97.750kΩ	97.750kΩ
R_D	2kΩ	1.9902kΩ	1.9902kΩ
R_S	2kΩ	2.0001kΩ	2.0001kΩ
R_{L1}	2kΩ	1.9791kΩ	1.9791kΩ
R_{L2}	2kΩ	1.9862kΩ	1.9862kΩ
V_{G1}	8.006V	7.6695V	7.8286V
V_{S1}	3.003V	2.7801V	2.8099V
V_{D1}	V	7.2199V	7.203V
I_{D1}	1.502mA	1.502mA	1.39mA

Procedure 3

INSERT PLOT AND TALK ABOUT IT.

Procedure 4

Using measurements from the resistors from the experiment the gain of the amplifier is calculated.

$$A_{CS} = 0.4961V/V$$

$$A_{SF} = 0.4895V/V$$

$$A_d = A_{SF} - A_{CS} = .9856$$

Using measurements from the voltages from the experiment the gain of the amplifier is calculated.

$$A_{CS} = \frac{v_{o1}}{v_{in}} = \frac{.459V}{1.024V} = 0.4482V/V$$

$$A_{SF} = \frac{v_{o2}}{v_{in}} = \frac{.462V}{1.024V} = 0.4512V/V$$

$$A_d = A_{SF} - A_{CS} = .8994$$

Calculating the gains of the circuit we can see the experimental setup has slightly less gain but has an error of $\epsilon = \frac{abs(1-.8994)}{1} = 10$ this is decent for the amplifier in a real lab with limited supply of resistors and mosfets.

Procedure 5

INSERT PICTURE FROM GOOGLE DRIVE AND WRITE ABOUT IT

Procedure 6

Parameter	Design Prediction	Adjusted Predictions	Actual Measurements
R_2	545Ω	545Ω	556Ω
R_3	667Ω	667Ω	659Ω
R_4	300Ω	300Ω	327Ω
V_{G2}	8.34V	8.4288V	8.3422V
V_{S2}	1.84V	1.9238V	2.3057V
V_{D2}	7.75V	7.7199V	8.0540V
I_D	3.38mA	3.38mA	3.46mA

After connecting the signal generator to the circuit the following values were recorded.

$$v_{in} = 0.992V$$

$$v_{s2} = 0.03526V$$

$$v_{d2} = 0.0784V$$

The gain of the circuit is calculated by $A_{CG} = \frac{v_{g1}}{v_{g2}} = \frac{7.8286V}{8.3422V} = 0.9384V/V$

PUT IN GRAPHS

WRITE CONCLUSION